

0	8	.	1
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A student prepared a plant root to observe cells undergoing mitosis.

He put the root in a small bottle of hydrochloric acid in a 40 °C water bath.

Why did he put the plant root in acid?

[1 mark]

0	8	.	2
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State **two** precautions required when working with hydrochloric acid.

[2 marks]

1 _____

2 _____

Question 8 continues on the next page

Turn over ►

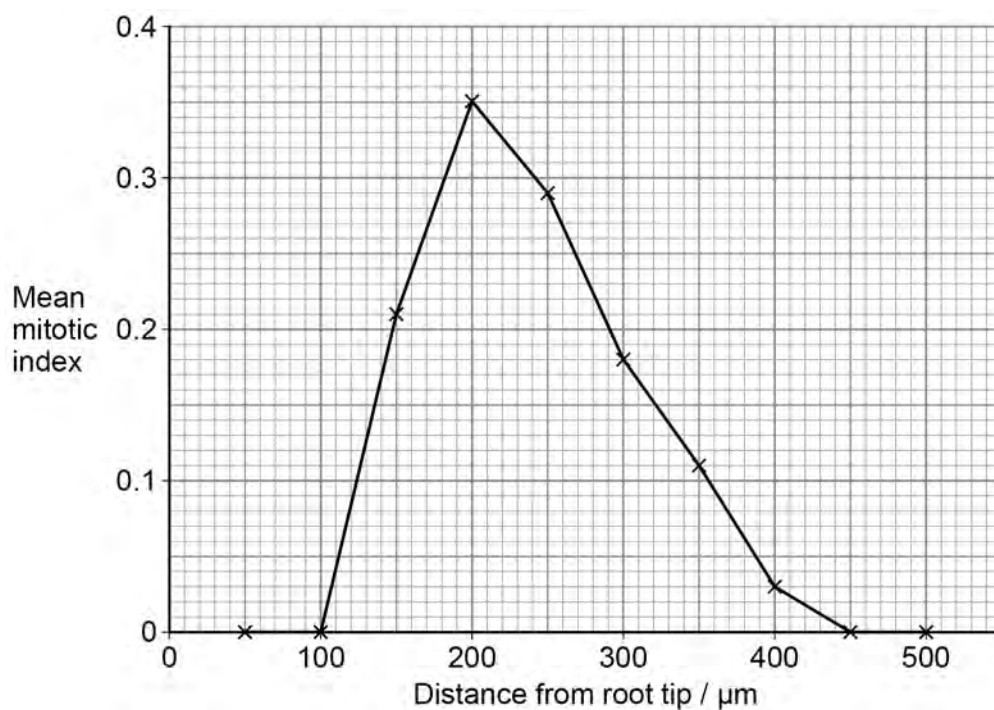


08.3

Scientists determined the mean mitotic index at 50 μm intervals away from the root tip in 10 young plant roots.

Figure 7 shows the scientists' results.

Figure 7



State the null hypothesis for this investigation.

Name the statistical test needed to determine whether the difference between the mean mitotic index at 200 μm and at 300 μm is significant.

[2 marks]

Null hypothesis _____

Statistical test _____



0 8 . 4

The scientists recorded these measurements from the tissue located at 200 μm from the root tip.

Area of field of view = 0.2 mm^2

Mean area of one cell = 3495 μm^2

Use this information and **Figure 7** to calculate the number of cells undergoing mitosis in this location.

Assume there are no spaces between the cells.

Show your working.

[2 marks]

_____ cells

0 8 . 5

Another student compares the mitotic index in the roots of two different species.

Give **two** considerations within her method to ensure this comparison is valid.

[2 marks]

1 _____

2 _____

9

Turn over for the next question

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